

Customer Churn EDA Report

1. Introduction

Customer churn refers to the loss of customers who stop using a company's products or services during a specific period. Understanding customer churn is important because retaining existing customers is generally less expensive than acquiring new ones. High churn rates can negatively affect revenue, profitability, and business growth. Therefore, analyzing churn patterns helps organizations identify the factors influencing customer departures and develop strategies to improve customer retention.

The objective of this project is to perform Exploratory Data Analysis (EDA) on a customer churn dataset to identify important churn patterns, understand customer behavior, and provide actionable business recommendations.

2. Dataset Details

Dataset Name: Customer Churn Dataset

Dataset Source: Kaggle

Number of Rows: 260,706

Number of Columns: 12

Target Variable: Churn

Dataset Features

Numerical Features

- Age
- Tenure
- Usage Frequency
- Support Calls
- Payment Delay
- Total Spend
- Last Interaction

Categorical Features

- Gender
- Subscription Type
- Contract Length

Identifier Column

- CustomerID

The target variable, **Churn**, indicates whether a customer has left the service (1) or remained active (0).

3. Data Cleaning Summary

Before conducting the analysis, several data cleaning steps were performed to improve data quality.

Missing Values

A small number of missing values were identified in multiple columns. Since the missing records represented only a tiny fraction of the dataset, they were removed using the `dropna()` method.

Duplicate Records

The dataset was checked for duplicate records to ensure that each customer was represented only once. Duplicate records were removed where necessary.

Data Type Corrections

The data types of all columns were verified to ensure that numerical and categorical variables were stored in appropriate formats for analysis.

Unnecessary Columns

The CustomerID column was removed because it serves only as a unique identifier and does not contribute to churn analysis.

Outlier Checks

Outliers were examined using boxplots, particularly for the Total Spend variable. Several high-spending customers were identified as outliers, but they were retained because they may represent valuable customer segments.

4. Exploratory Data Analysis

Several visualizations and statistical techniques were used to explore customer behavior and churn patterns.

Churn Distribution

A count plot and pie chart were created to analyze customer churn distribution.

Observation: Approximately 95.89% of customers churned, while only 4.11% remained active.

Age Distribution

A histogram was used to study the age distribution of customers.

Observation: Customers belong to a wide range of age groups, indicating a diverse customer base.

Tenure Distribution

A histogram was used to examine customer tenure.

Observation: The dataset contains both short-term and long-term customers, making tenure an important variable for churn analysis.

Total Spend Analysis

A boxplot was used to detect spending outliers.

Observation: Several customers have significantly higher spending than average, suggesting the presence of premium customer segments.

Subscription Type vs Churn

A count plot was used to compare churn across subscription plans.

Observation: Customers with Basic subscriptions showed the highest churn rate.

Contract Length vs Churn

A count plot was used to compare churn across contract durations.

Observation: Monthly contract customers exhibited the highest churn rate.

Correlation Analysis

A correlation heatmap was used to examine relationships among numerical variables.

Observation: Support Calls and Payment Delay showed positive relationships with churn, while Total Spend and Tenure showed negative relationships with churn.

5. Key Findings

1. The overall customer churn rate is approximately **95.89%**.
 2. Basic subscription customers exhibit the highest churn rate.
 3. Monthly contract customers show the highest likelihood of churn.
 4. Support Calls have the strongest positive relationship with churn.
 5. Payment Delay is positively associated with customer churn.
 6. Customers with higher Total Spend are less likely to churn.
 7. Longer customer tenure is associated with improved retention.
 8. Contract length, customer support interactions, and spending behavior are major factors influencing churn.
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6. Business Recommendations

1. Promote Long-Term Contracts

Encourage customers to choose annual or quarterly plans by offering discounts, rewards, and exclusive benefits.

2. Improve Customer Support

Enhance customer support quality and reduce issue resolution times to decrease customer dissatisfaction.

3. Reduce Payment Delays

Implement automated payment reminders and flexible payment options to minimize billing-related churn.

4. Strengthen Customer Retention Programs

Introduce loyalty programs, personalized offers, and customer engagement initiatives.

5. Monitor High-Risk Customers

Identify customers with frequent support calls, payment delays, and low spending levels, and proactively engage them through retention campaigns.

6. Increase Customer Engagement

Encourage product usage through targeted communication, educational resources, and promotional campaigns.

7. Conclusion

This project conducted an exploratory data analysis of a customer churn dataset containing more than 260,000 customer records. Data cleaning, statistical analysis, and visualization techniques were applied to identify important factors associated with customer churn.

The analysis revealed a very high churn rate of approximately 95.89%. Contract length, support calls, payment delays, customer spending behavior, and tenure were identified as key factors influencing churn. Monthly contract customers and customers requiring frequent support were found to be at the highest risk of leaving.

Based on these findings, businesses can improve customer retention by enhancing customer support services, encouraging long-term contracts, reducing payment-related issues, and implementing targeted retention strategies. These actions can help improve customer satisfaction, increase loyalty, and support long-term business growth.